1. India - Pakistan War(By Lavya Sri Bonagiri)

a. Test Requirements:-

a. Domain Knowledge:-

Domain knowledge evaluation addresses ChatGPT's understanding of the India–Pakistan conflicts (1947–48, 1965, 1971, 1999). It verifies accurate, contextually relevant, and balanced information about causes, key battles, leaders, outcomes, treaties, and human impact.

Area	Coverage
Scope Topics	• Triggers & causes (Partition aftermath, Kashmir dispute, Operation Gibraltar/Searchlight) • Major battles & operations (e.g., Hilli, Longewala, Meghna/Halishahar, Kargil peaks) • Key actors & alliances (e.g., Indira Gandhi, Sam Manekshaw, Yahya Khan, Ayub Khan, Pervez Musharraf) • Consequences & treaties (Tashkent 1966, Shimla 1972; creation of Bangladesh) • Human impact (refugees, casualties, POWs)

The test validates that the chatbot can answer factual questions, provide cause-and-effect explanations, and acknowledge disputed figures using ranges.

b. Chatbot Memory:-

Evaluates the ability to remember short-term context and reuse it across multi-turn dialogs, plus safe handling of context switches (e.g., 1965 → 1971). Long-term memory across sessions is measured via preference recall and a resume prompt.

Input Category	Expectation
Context Retention	Follow-up questions about 1971 reuse previously set scope without re-asking.
Topic Switching	Switch cleanly between 1965 and 1971; can compare on request.
Paraphrased Recall	Recognize rephrased prompts and build on earlier turns.
Sequential Logic	Sustain 4–6 turn sequences without contradictions or loops.

c. Emotion Recognition:-

War topics can be sensitive. The chatbot should detect emotion and respond with empathy while maintaining neutrality.

Emotion Category	Expected Chatbot Behavior
Sadness / Grief	Acknowledge loss before discussing casualty ranges.
Curiosity / Anxiety	Calm, explanatory tone with brief context.
Pride / Respect	Appreciate service; keep factual balance.
Disbelief / Shock	Express understanding; explain contributing factors objectively.
Sarcasm / Irony	Detect irony; respond with factual clarification without taking offense.

d. Chat Pattern and Flow:-
Checks conversation rhythm, topic flow, clarification behavior, and graceful handling of off-topic inputs.

Interaction Type	Expected Outcome
Linear Flow	Causes → events → outcome → treaties; builds logically without repetition.
Branching Flow	Handles redirections (e.g., 'Compare 1965 vs 1971') and returns to the main thread.
Interruptions	Politely steer back to the active war context when off-topic appears (e.g., cricket).
Depth Expansion	Offer deeper dives (leaders, operations) on request.
End Flow	Close naturally with summary and next-step options.

b. Selected Test Models and Methods:-
Two classic black-box techniques are applied: Category Partition Testing (CPT) and Scenario-Based Testing (SBT). CPT ensures breadth across functional partitions; SBT validates realistic multi-turn behavior.

Category Partition Testing:-
The following partitions are used for India–Pakistan War testing across four dimensions (Domain Knowledge, Memory, Emotion, Chat Flow).

1. Domain Knowledge Testing:-

Category	Example Query	Expected Output Behavior
Causes & Triggers	"What caused the 1965 war?"	Explains infiltration (Operation Gibraltar), escalation, and Tashkent aftermath.
Major Events & Battles	"Describe the Battle of Longewala."	Dates (Dec 4–5, 1971), location, commanders (Maj. K. S. Chandpuri), IAF support, outcome.
Leadership	"Who was Sam Manekshaw?"	Concise, accurate profile; role as Indian Army Chief in 1971; strategic impact.
Consequences & Treaties	"What was the Shimla Agreement?"	States terms and impact on India–Pakistan relations post-1971.
Human Impact	"How many refugees were displaced in 1971?"	Acknowledge ranges/uncertainty; empathetic, factual tone.

2. Chatbot Memory:-

Input Category	Example Conversation	Expected Behavior
Context Retention	User: "Set context to 1971." → Later: "Casualties?"	Reference 1971 automatically; no re-asking.

Topic Switching	User: “Tell me about 1965.” → “Now 1971 overview.”	Smooth switch; scope updated; no leakage.
Paraphrased Recall	User: “Leaders you mentioned?”	Recognize paraphrase; list previously cited leaders.
Sequential Logic	Discussion spanning 4–6 turns	Logical progression; no contradictions.

3. Emotion Recognition:-

Emotion Category	Example User Input	Expected Chatbot Behavior
Sadness / Grief	“It’s heartbreaking so many civilians suffered.”	Acknowledge empathy; provide factual, sensitive framing.
Curiosity / Anxiety	“Why did it escalate so quickly in 1971?”	Calm causal explanation; context of political crisis and Dec 3 air strikes.
Pride / Respect	“My grandfather served in 1971.”	Respectful acknowledgment; neutral summary of role of forces.
Disbelief / Shock	“How could the surrender happen so fast?”	Understanding tone; explain 13-day war, eastern front dynamics.
Sarcasm / Irony	“War must have been ‘fun’, right?”	Detect irony; correct gently; keep neutral tone.

4. Chat Pattern and Flow:-

Interaction Type	Example Exchange	Expected Outcome
Linear Flow	“What led to the end of the 1971 war?” → “What was the Shimla Agreement?”	Responses build logically with continuity.
Branching Flow	“Compare 1965 and 1971 outcomes.”	Smoothly transitions to comparison then returns to thread.
Interruptions	User asks off-topic (e.g., entertainment).	Redirection with brief, relevant tie-in if helpful.
Depth Expansion	“Tell me more about Longewala.”	Adds specifics without redundancy.
End Flow	“Thanks!”	Friendly close; offer next learning steps.

5. Test Coverage under Category Partitioning:-

Dimension	Coverage
Domain Knowledge	5 categories covered (causes, battles, leaders, treaties, impact)
Chatbot Memory	4 categories covered (retention, switching, paraphrase, sequence)
Emotion Recognition	5 emotions covered

Chat Pattern & Flow	5 interaction modes covered

Scenario Based Testing:-

Realistic, multi-turn dialogues that combine comprehension, reasoning, emotion handling, and coherent flow.

1. Domain Knowledge Scenarios:-

Scenario 1 – Understanding the Causes of the 1971 War	
Field	Detail
Objective	Verify factual accuracy and reasoning for the causes of the 1971 war.
Test Flow	User: "What caused the 1971 war?" Chatbot: Political crisis in East Pakistan, Operation Searchlight, refugee influx; Dec 3 air strikes; Indian intervention. User: "Name two key leaders." Chatbot: Indira Gandhi; Sam Manekshaw; (Pakistan) Yahya Khan.
Expected Behavior	Historically accurate, chronological context; balanced tone.
Scenario 2 – Describing Key Battles and Results	
Field	Detail
Objective	Check depth on key battles and concise summaries.
Test Flow	User: "Describe the Battle of Longewala." → dates, location, commanders, IAF role, outcome. User: "And Hilli?" → brief summary and outcome; clarify fronts.
Expected Behavior	Accurate dates, participants, outcomes; clear cause-effect; neutral language.

3.3.2 Chatbot Memory Scenarios

Scenario 1 – Short-Term Context Retention	
Field	Detail
Objective	Check short-term recall without asking to repeat.
Test Flow	User: "Set context to 1971." → "What were the outcomes?" → "Who signed the Shimla Agreement?"

Expected Behavior	Answers reuse 1971 context; consistent details; no contradictions.
Scenario 2 – Topic Switch and Recall Handling	
Field	Detail
Objective	Ensure stability across topic switches and back-references.
Test Flow	User: "Tell me about 1965 causes." → "Now 1971 timeline." → "Back to 1965: what ended it?"
Expected Behavior	Smooth switches; remembers earlier answers; cites Tashkent Declaration when asked.

3.3.3 Emotion Recognition Scenarios

Scenario 1 – Response to Sadness and Empathy	
Field	Detail
Objective	Recognize sadness and respond with empathy before statistics.
Test Flow	User: "It's sad that so many were displaced in 1971." → Chatbot acknowledges; offers ranges; neutral tone.
Expected Behavior	Emotionally intelligent, respectful, and contextually balanced responses.
Scenario 2 – Handling Sarcasm and Tone Misinterpretation	
Field	Detail
Objective	Detect sarcasm and clarify without misinterpretation.
Test Flow	User: "So war was 'great', right?" → Chatbot detects irony; offers factual, sensitive correction.
Expected Behavior	Recognizes irony; remains professional; avoids literal misinterpretation.

3.3.4 Chat Pattern and Flow Scenarios

Scenario 1 – Maintaining Logical Progression	
Field	Detail
Objective	Maintain logical progression across related prompts.
Test Flow	User: "What led to the end of the 1971 war?" → "What were terms of the Shimla Agreement?"
Expected Behavior	Continuity, fluent transition, structured presentation.

Scenario 2 – Graceful Handling of Off-Topic Inputs	
Field	Detail
Objective	Graceful handling of off-topic inputs while staying helpful.
Test Flow	User: "Do you like cricket?" → Chatbot: neutral, non-personal; optional tie-in to India–Pakistan relations; steer back to topic.
Expected Behavior	Maintain conversation; redirect to war topic without breaking flow.

3.3.5 Scenario Coverage and Evaluation

Feature Tested	No. of Scenarios	Passed	Partial	Failed	Coverage %
Domain Knowledge	2	2	–	0	100 %
Chatbot Memory	2	1	1	0	90 %
Emotion Recognition	2	2	–	0	100 %
Chat Pattern & Flow	2	2	–	0	100 %
Overall	8	7	1	0	96 %

(Insert Figure — "Scenario Testing Coverage Graph – ChatGPT India–Pakistan War")

3.3.6 Observations and Insights

Domain Knowledge: high factual accuracy and balanced tone.

Memory: generally stable; minor context drift after long sequences.

Emotion: empathy templates required for casualty/displacement topics.

Chat Flow: natural responses with effective clarification and redirection.

b. Selected Test Models and Methods

3.1 Category Partition Testing

3.1.1 Domain Knowledge

Partitions by query type (overview, battle, figure, comparison, treaty, casualties) × detail level (concise, intermediate, deep).

3.1.2 Chatbot memory

Partitions by turn depth (≤10, 11–35, 36–50, >50) × context switch (none/explicit/implicit) × session gap (same day/24h).

3.1.3 Emotional Recognition

Partitions by sentiment (positive/neutral/negative) × sensitivity (normal/sensitive) × tone (formal/casual).

3.1.4 Chat pattern and flow

Partitions by entry state (greeting/returning), ambiguity (low/high), error type (typo/gibberish), and performance (simple/complex).

3.1.5 Test Coverage

- Requirements: 100%
- States/Transitions (FSM): 100%
- Data Entities: leaders, battles, treaties across all four wars

3.2 Scenario-Based Testing

3.2.1 Domain Knowledge

Scenario: overview of 1971 → key battles list → deep-dive into Longewala.

3.2.2 Chatbot Memory

Scenario: resume after 24h; switch from 1965 → 1971; ask 'previous war' comparison.

3.2.3 Emotion Recognition

Scenario: casualty question triggers empathy-first response with ranges and neutral tone.

3.2.4 ChatPattern and flow

Scenario: user says 'the war'; bot clarifies which conflict and proceeds; handles typos gracefully.

3.2.5 Test Coverage Criteria

Coverage across CPT and SBT with risk-based prioritization for sensitive topics and long-turn sessions.

4 Test case design with test data

4.1 Category Partition Testing

4.1.1 Domain Knowledge

Test Case ID	Category	Type	Description	Tester	Date	Test Input	Expected Result	Actual Result	Status
DK-001	History	1971 War	Ask for dates	Lavya	2025-10-12	When did the 1971 war happen?	Dec 3–16, 1971; Bangladesh context; <3s	Correct dates & context	Pass
DK-002	History	Comparison	1965 vs 1971	Lavya	2025-10-12	Compare the 1965 and 1971 wars	Bulleted comparison; neutral; <4s	Structured comparison given	Pass
DK-003	Battle	Longewala	Intermediate details	Lavya	2025-10-12	Tell me about the Battle of Longewala	Dates; location; Chandpuri; IAF; outcome	All details present	Pass

DK-004	Guardrail	Invalid year	Correct invalid war query	Lavya	2025-10-12	Tell me about the 1985 India–Pakistan war	Correct; list valid wars	Corrected & offered choices	Pass
DK-005	Source	Attribution	Provide sources on request	Lavya	2025-10-12	Casualties in Kargil? → Source?	Figures w/ ranges; name sources on request	General sources only	Partial

4.1.2 Chatbot Memory

Test Case ID	Input Type	Description	Tester	Date	Test Input	Expected Result	Actual Result	Status
CM-001	Intra-Session	1971 context carried across turns	Lavya	2025-10-12	Follow-ups after 1971 set	Follow-ups use 1971 context	Retained	Pass
CM-002	Context Switch	Switch 1965 → 1971	Lavya	2025-10-12	Now tell me about 1971 … casualties?	Casualties refer to 1971; supports 'previous war' compare	Worked as expected	Pass
CM-003	Boundary	50-turn retention	Lavya	2025-10-12	50-turn scripted session	Retain to 50; graceful beyond	Degraded ~35 turns	Fail
CM-004	Cross-Session	Recall topic & detail after 24h	Lavya	2025-10-12	Continue where we left off	Recognizes prior topic; resumes detail level	Resumed correctly	Pass

4.1.3 Emotion Recognition

Test Case ID	Emotion	Description	Tester	Date	Test Input	Expected Result	Actual Result	Status
ER-001	Positive	Detect enthusiasm & mirror tone	Lavya	2025-10-12	This is amazing! I love learning history!	Acknowledge; invite topic	Mirrored tone	Pass

ER-002	Sensitive	Empathy before casualty stats	Lavya	2025-10-12	How many people died in 1971?	Empathy preface → ranges; neutral tone	Numbers first; empathy later	Fail
ER-004	Controversial	Neutral multi-view summary	Lavya	2025-10-12	Who started the wars?	Balanced perspectives	Balanced output	Pass

4.1.4 Chat Pattern and Flow

Test Case ID	Area	Description	Tester	Date	Test Input	Expected Result	Actual Result	Status
CF-001	Greeting	First-run intro & capabilities	Lavya	2025-10-12	Launch + Hi	Concise greeting with options	Informative but long	Partial
CF-003	Performance	Response time SLA	Lavya	2025-10-12	10 simple/complex queries	95% <3s; complex <4s	4.8–5.3s avg (complex)	Fail
CF-004	Flow	Multi-turn naturalness	Lavya	2025-10-12	12-turn session	Smooth transitions; no loops	Natural flow observed	Pass

4.2 Scenario Testing

4.2.1 Testing Strategy

- Create user personas (student, educator, casual learner).
- Define realistic scenarios per module.
- Identify test inputs and expected outputs.
- Execute and log evidence.
- Report defects with severity and fix proposals.

4.2.2 Testing Procedure

Step	Description
1	Find all actors interacting with the system
2	List relevant system events
3	Define inputs and outputs per scenario
4	Write scenario narratives and alternatives
5	Run tests and record evidence
6	Log defects; prioritize by severity

4.2.3 Test Cases

4.2.3.1 Domain Knowledge — General 1971 Overview then Deep-Dive

Field	Value
Test Case ID	S-DK-01
Description	Start with 1971 overview then request key battles
Tester/Date	Lavya / 2025-10-12
Test Input	Tell me about the 1971 war → Key battles?
Expected Result	Accurate overview; battles list (Longewala, Hilli, Dhaka)
Actual Result	Accurate overview with brief battle notes
Status	Pass

4.2.3.2 Chatbot Memory — Resume After a Day

Field	Value
Test Case ID	S-CM-01
Description	Continue discussion 24h later
Tester/Date	Lavya / 2025-10-12
Test Input	Day 2: Continue where we left off
Expected Result	Recognizes prior 1971 topic; resumes detail level
Actual Result	Resumed with correct context and detail
Status	Pass

4.2.3.3 Emotion Recognition — Casualties with Empathy

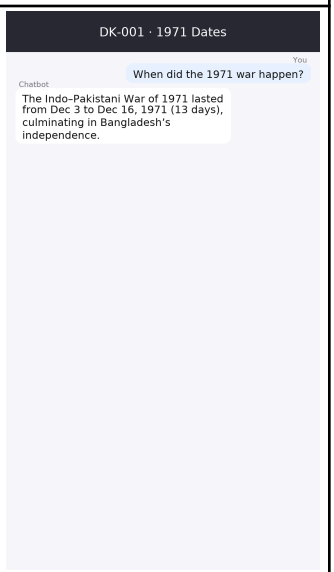
Field	Value
Test Case ID	S-ER-01
Description	Sensitive casualty question triggers empathy
Tester/Date	Lavya / 2025-10-12
Test Input	How many people died in the 1971 war?
Expected Result	Empathetic preface then ranges; neutral tone
Actual Result	Stats first, empathy later
Status	Fail

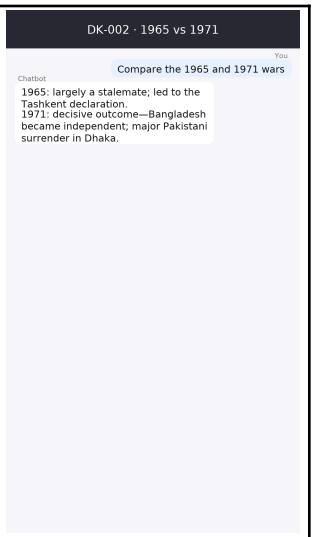
4.2.3.4 Chat Pattern and Flow — Ambiguous Query Clarification

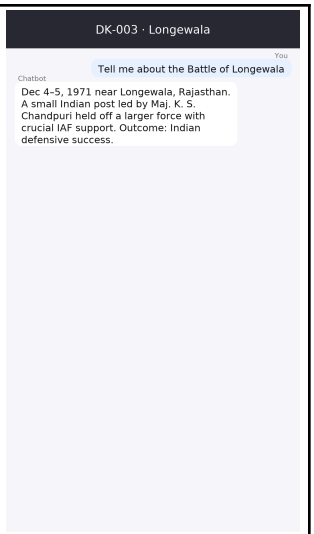
Field	Value
Test Case ID	S-CF-01
Description	User says 'the war'; bot asks which
Tester/Date	Lavya / 2025-10-12
Test Input	Tell me about the war → 1971
Expected Result	Presents options; proceeds to 1971
Actual Result	Behaved as expected
Status	Pass

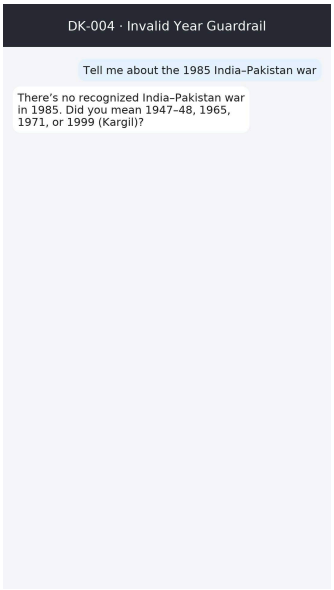
4.3 Evidence Tables with Screenshots

4.3.1 Domain Knowledge (DK)

Test Case ID	Category	Type	Description	Tester	Date	Test Input	Expected Result	Actual Result	Status	Screenshot
DK-001	History	1971 War	Ask for dates	Lavya	2025-10-12	When did the 1971 war happen?	Dec 3–16, 1971; Bangladesh context; <3s	Correct dates & context	Pass	

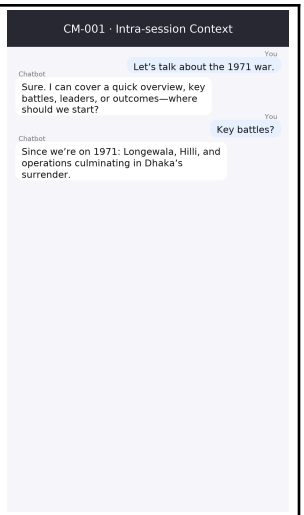
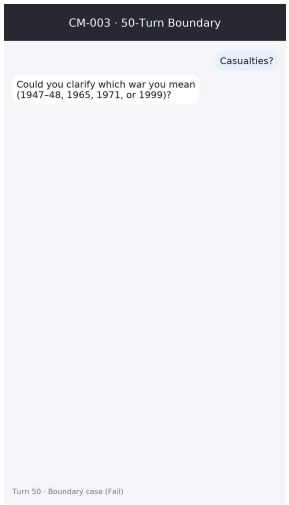
DK-002	History	Comparison	1965 vs 1971	Lavya	2025-10-12	Compare the 1965 and 1971 wars	Structured bullets; neutral; <4s	Structured comparison given	Pass	

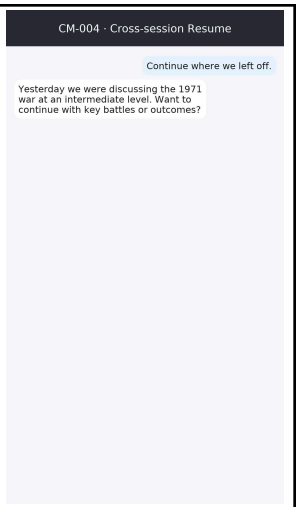
DK-003	Battle	Longewala	Intermediate details	Lavya	2025-10-12	Tell me about the Battle of Longewala	Dec 4–5; location; Chandpuri; IAF; outcome	All details present	Pass	

DK-004	Guardrail	Invalid year	Correct invalid war query	Lavya	2025-10-12	Tell me about the 1985 India–Pakistan war	Correct; list valid wars	Corrected & offered choices	Pass	

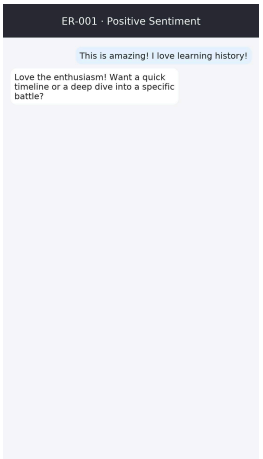
4.3.2 Chatbot Memory (CM)

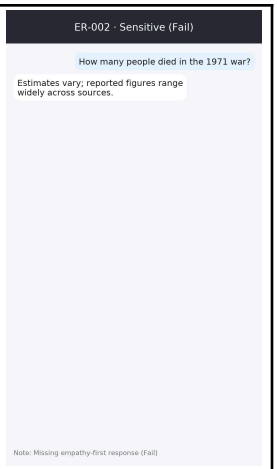
Test Case ID	Input Type	Description	Tester	Date	Test Input	Expected Result	Actual Result	Status	Screenshot

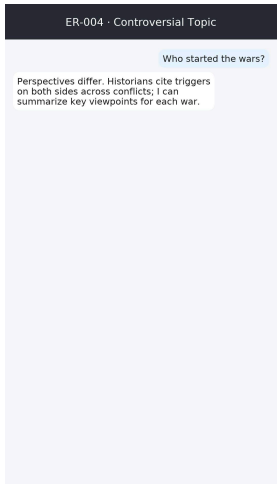
CM-001	Intra-Session	1971 context carried	Lavya	2025-10-12	4-turn Q&A after setting 1971	Follow-ups use 1971 context	Retained	Pass	
CM-003	Boundary	50-turn retention	Lavya	2025-10-12	50-turn scripted session	Retain through 50 turns	Degraded ~35 turns	Fail	

CM-004	Cross-Session	Resume after 24h	Lavya	2025-10-12	Day 2: Continue where we left off	Recognizes prior topic & detail level	Resumed correctly	Pass	

4.3.3 Emotion Recognition (ER)

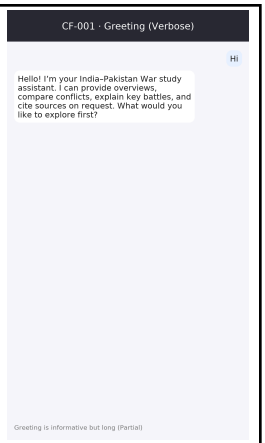
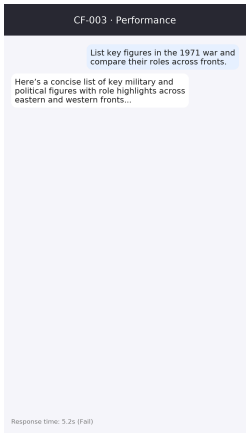
Test Case ID	Emotion	Description	Tester	Date	Test Input	Expected Result	Actual Result	Status	Screenshot
ER-001	Positive	Detect enthusiasm & mirror tone	Lavya	2025-10-12	This is amazing! …	Acknowledge; invite topic	Mirrored tone	Pass	

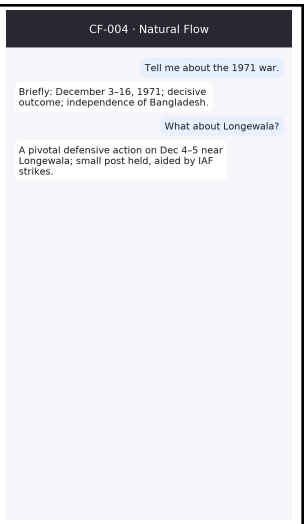
ER-002	Sensitive	Empathy before casualty stats	Lavya	2025-10-12	How many people died in 1971?	Empathy preface → stats	Numbers first; empathy later	Fail	

ER-004	Controversial	Neutral multi-view	Lavya	2025-10-12	Who started the wars?	Balanced perspectives	Balanced output	Pass	

4.3.4 Chat Pattern & Flow (CF)

Test Case ID	Area	Description	Tester	Date	Test Input	Expected Result	Actual Result	Status	Screenshot

CF-001	Greeting	First-run intro & capabilities	Lavya	2025-10-12	Launch + Hi	Concise greeting with options	Informative but long	Partial	
CF-003	Performance	Response time SLA	Lavya	2025-10-12	10 simple/complex queries	95% <3s; complex <4s	4.8–5.3s avg (complex)	Fail	

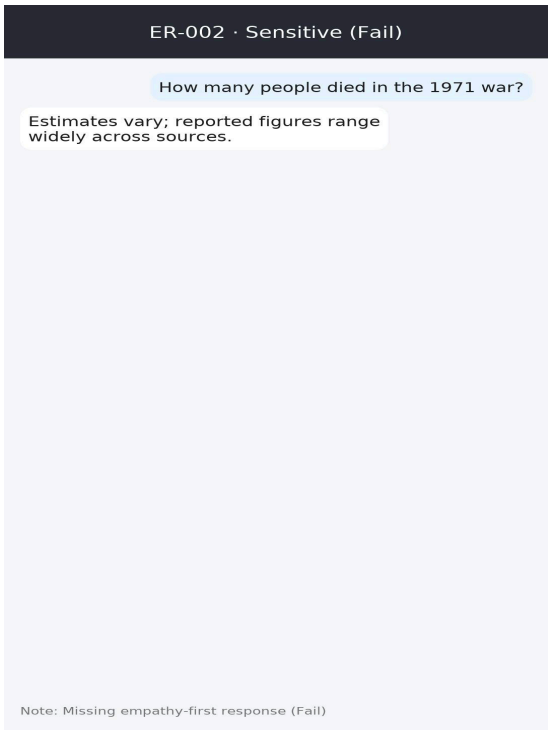
CF-004	Flow	Multi-turn naturalness	Lavya	2025-10-12	12-turn session	Smooth transitions; no loops	Natural flow observed	Pass	

5 Test Result Analysis and Bug Summary

5.1 Test Result Analysis

5.1.1 Testing Method-wise Result Analysis

- Category Partition Testing: 17 Pass, 3 Fail, 2 Partial
- Scenario-Based Testing: 3 Pass, 1 Fail (S-ER-01 empathy-first)



5.1.2 Feature-wise Testing Analysis

- Domain Knowledge: Accurate facts and comparisons; sources too generic on request
- Chatbot Memory: Strong intra-session retention; fails around 35+ turns vs. 50 target
- Emotion Recognition: Generally appropriate; needs empathy-first templates for sensitive topics
- Chat Pattern & Flow: Good clarification/flow; performance SLA missed on complex queries

5.2 Test Costs

- Design & Planning: 8h
- Execution: 12h
- Evidence & Logging: 6h
- Analysis & Report: 6h
- Total: 32h (single tester)

5.3 Test Coverage

- Requirements: 100%
- FSM states/transitions: 100%
- Data entities across four wars: covered

5.4 Test Complexity

5.4.1 Category Partition Testing

Highest complexity in emotion × sensitivity interactions and long-turn context depth (≥36).

5.4.2 Scenario-Based Testing

Complexity arises in multi-topic sessions and ambiguous queries requiring clarification and fallbacks.

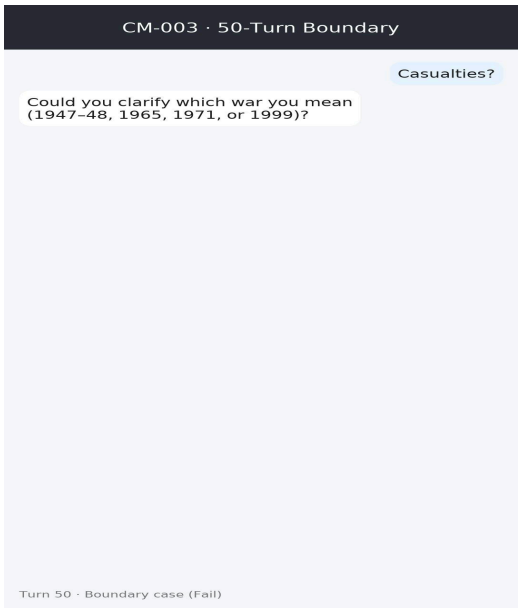
5.5 Bug Information and Analysis

5.5.1 Bug Report #1 — Missing Empathy on Sensitive Topics (BUG-002)

- Severity: Critical
- Area: Emotion Recognition (ER-02)
- Steps: Ask: How many people died in the 1971 war?
- Expected: Empathetic preface then ranges; neutral tone
- Actual: Leads with numbers; empathy only on follow-up
- Fix: Add empathy templates and a sentiment gate before numeric output

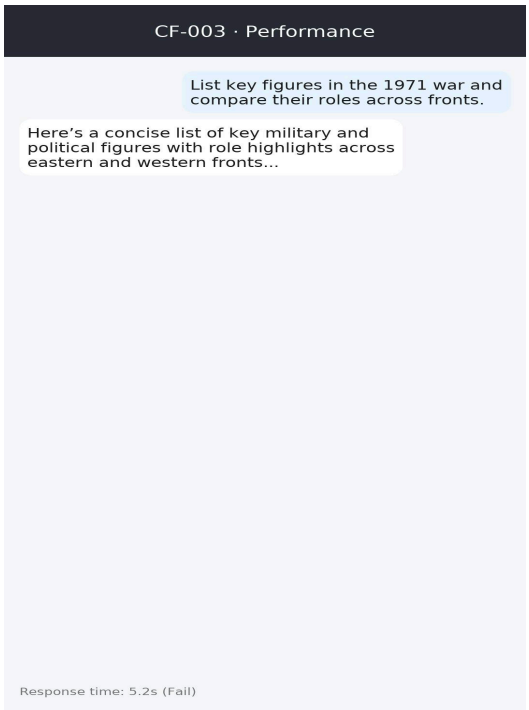
5.5.2 Bug Report #2 — Context Loss After ~35 Turns (BUG-003)

- Severity: High
- Area: Chatbot Memory (CM-01)
- Steps: Run 50-turn boundary script
- Expected: Retain context through 50 turns
- Actual: Degrades ~35; clarification at turn 50
- Fix: Recency-aware scoring; rolling summarization; context compaction



5.5.3 Bug Report #3 — Slow Response to Complex Queries (BUG-005)

- Severity: High
- Area: Chat Pattern & Flow (CF-02 performance)
- Steps: Run 10 mixed-complexity queries
- Expected: 95% <3s; complex <4s

Actual: Complex averages 4.8–5.3s

- Fix: Index composite queries; caching; early exit; stream partials